

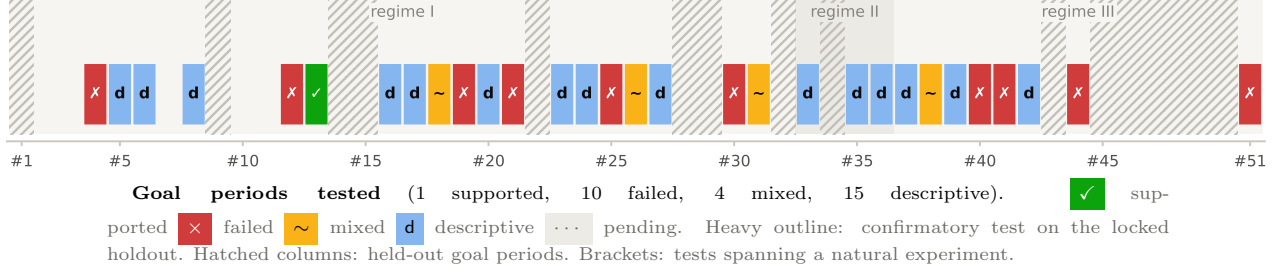
H53: Herding is announcement-seeded nucleation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

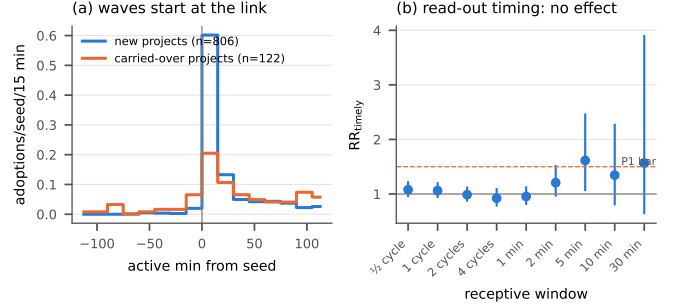
Bottom line. Herding waves are seeded by announcements and bounded by rooms, but their size is set by the project's existing share and the live conversation, not by how many agents read the link within one call cycle; the call clock only binds for deadline-bound calls.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Is a herding wave's size set by how many uncommitted agents read the seeding link within one model-call cycle? We took the first chat link to each project (928 seeds, 23 non-holdout periods). Read-out timing came from the context ledger. We validated on synthetic swarms run on the real call schedules first. *Seeding holds*: adoptions jump ~ 20 -fold at the first link, stay in the poster's room (other-room ratio 0.12), and almost never happen without the link in context (1.5%). *The clock fails*: prompt readers adopt no more often than later readers ($RR_{\text{timely}} = 1.06 [0.93, 1.22]$, 726 adoptions). The receptive count adds nothing to held-out wave-size prediction; the project's current share is the only predictor (AUC 0.72 vs 0.46). Poster status adds nothing. The exception is deadline-bound calls: in the #26 runoff, ballots = readers inside the 100-s window ($8 = 8$). Holdout not run.



Observables. (a) Adoptions per seed in 15-min active bins around a project's first chat link. New projects: almost nothing before, a spike after. Carried-over projects: a smaller step ($3.8\times$). (b) Adoption within 60 min of an agent's own read-out, early vs late readers, for receptive windows from half a call cycle (8 s) to 30 min. Flat at the cycle scale; noisy at the minutes scale. Dashed: the pre-registered bar of 1.5.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful, mostly as a negative. Operators can stop trying to time project announcements to the agents' call loops: posting when many agents are "about to read" buys about 9%. What does move a pile-on is whether there is a link at all (no link, almost no wave), whether the room already has agents on the project, and whether readers are in the live conversation ($3\times$). For deadline-bound calls such as votes, polls or sign-ups, the clock binds: the wave equals the agents whose next call lands inside the window. In regime III a 2-min window misses about a third of the swarm; ≥ 15 min reaches about 90%. That is a concrete design rule, although it rests on one election.

PHYSICAL MODEL

Models 10 and 03: kinetic Potts with read-out-gated updates. Agent i (state σ_i : project or none) can switch only at its own model calls. A seed (first chat link to project X) is a field pulse felt at i 's reading call, at message age a_i . A_X is attractiveness, u_i uncommitted, c_i i 's call cycle, g a decreasing gate, k_X current adopters. R is the receptive count. Rivals: poster status, a shared field, room size N , current share.

$$P(\sigma_i \rightarrow X | a_i) = 1 - \exp[-A_X u_i g(a_i/c_i) e^{\beta J k_X / (N-1)}]$$

$$\mathbb{E}[S] \approx A_X [p_{\text{hi}} R + p_{\text{lo}} (N_{\text{sus}} - R)], \quad R = \#\{i : u_i = 1, a_i \leq c_i\}$$

$$\log \mathbb{E}[y_{is}] = \alpha_s + \delta \mathbf{1}[a_i \leq c_i] + \gamma u_i + \dots, \quad RR_{\text{timely}} = e^\delta$$

RESULTS VS PREDICTIONS

Prediction	Observed
P1 $RR_{\text{timely}} \geq 1.5$	1.06 [0.93, 1.22]
P1b 10-min window	1.35 [0.79, 2.29]
P2 R wins (held-out)	-1.7 nats vs M0; share +62
P3 status minor	-1.4 nats vs M0
P4 $AUC(R) \geq 0.65$	0.46 (share 0.72)
P6 no shared field	step 22.5; other room 0.12
P9 timing gain	$\times 1.09$
#26 read-out gating	17/17 after read-out
#26 runoff wave	8 ballots = 8 readers
#40 hub = field	link-seeded; $R = 0$

Verdict

$\times$
 $\times$
 $\times$
 $\checkmark$
 $\times$
 $\checkmark$
 $\times$
 $\checkmark$
 $\checkmark$
 $\checkmark$
 $\times$

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

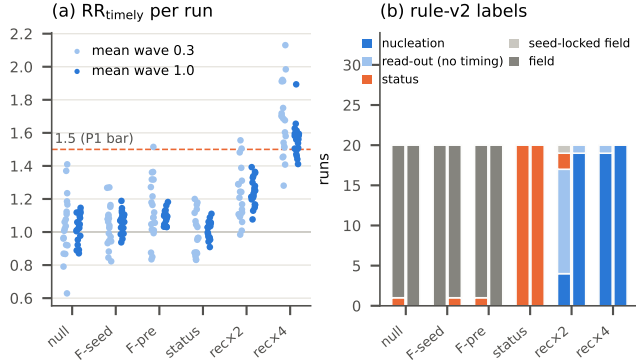
Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Adoption is attention (strict artifact mentions); action-only labels agree, and commit waves are rare. The step at the first link is partly mechanical for brand-new projects ($3.8\times$ for carried-over ones). “Uncommitted” is a crude proxy. The agent-level test detects $\times 4$ receptive effects, not reliably $\times 2$, and the seed-level test had no power. The seed models’ partial pooling is post hoc (Amendment 2). In #26 the recorded approval opening came after most ballots.

ROUND 2

- **Essence**. When does announcing something produce a pile-on, and can an operator shape the size by when and how it announces?
- **H53-R1** Deadline-bound calls as the regime where the clock binds.
- **H53-R2** Conversation-gated uptake.
- **H53-R3** Within-project designs at scale.
- **H53-R4** A better commitment measure.



Second figure. Synthetic validation on the real schedules (20 runs per world and base rate). (a) The agent-level test separates planted receptive nucleation ($\times 2$, $\times 4$) from activity, field and status worlds. (b) The frozen decision rule: no false “nucleation” in 120 non-nucleation runs; $\times 2$ is recovered only at the higher base rate.

GOAL PERIODS IN DETAIL

G04 $\times$ failed — RR 0.69 [0.47, 1.01]; $F1 \infty$ (no pre-seed adoption); mean S ...
 G05 d descriptive — 6 seeds; mean S 0.67
 G06 d descriptive — 7 seeds; mean S 0.43
 G08 d descriptive — RR 0.32 [0.03, 2.88]; $F1 \infty$ (no pre-seed adoption); me...
 G12 $\times$ failed — RR 1.39 [0.80, 2.43]; $F1 \infty$ (no pre-seed adoption); mean S ...
 G13 $\checkmark$ supported — RR 1.92 [1.15, 3.23]; $F1$ 11.50; mean S 0.82; R 1.48
 G16 d descriptive — 8 seeds; mean S 0.50
 G17 d descriptive — RR 1.26 [0.37, 4.23]; $F1 \infty$ (no pre-seed adoption); me...
 G18 $\sim$ mixed — RR 1.38 [1.03, 1.83]; $F1$ 101.00; mean S 2.21; R 1.44
 G19 $\times$ failed — RR 1.39 [0.76, 2.54]; $F1$ 30.00; mean S 0.71; R 1.10
 G20 d descriptive — RR 0.56 [0.54, 0.57]; $F1 \infty$ (no pre-seed adoption); me...
 G21 $\times$ failed — RR 0.76 [0.42, 1.36]; $F1 \infty$ (no pre-seed adoption); mean S ...
 G23 d descriptive — 5 seeds; mean S 2.20
 G24 d descriptive — RR 0.56 [0.56, 0.56]; $F1 \infty$ (no pre-seed adoption); me...
 +16 more periods: see the card.